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Analyzing Cross-Trait Genetic Architecture with the BIGA Cloud Computing Platform

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ABSTRACT

As large-scale biobanks provide increasing access to deep phenotyping and genomic data, genome-wide association studies (GWAS) are rapidly uncovering the genetic architecture behind various complex traits and diseases. GWAS publications typically make their summary-level data (GWAS summary statistics) publicly available, enabling further exploration of genetic overlaps between phenotypes gathered from different studies and cohorts. However, systematically analyzing high-dimensional GWAS summary statistics for thousands of phenotypes can be both logistically challenging and computationally demanding. In this article, we introduce BIGA (https://bigagwas.org/), a website that aims to offer unified data analysis pipelines and processed data resources for cross-trait genetic architecture analyses. We have curated over 15,000 phenotypes and integrated four statistical genetics tools to support cloud-based batch-running for various cross-trait data analyses. Additionally, we developed a novel online query function for GWAS summary statistics, linking to over 30,000 harmonized datasets in the GWAS Catalog. Through BIGA, users can upload their own data, query data online without local downloads, submit jobs, and share results, providing the research community with a convenient tool for consolidating GWAS data and generating new insights. Supplementary materials for this article are available online, including a standardized description of the materials available for reproducing the work.

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Cloud computing; Cross-trait analysis; GWAS; Online platform

1. Introduction

The rapid development of biobank-scale biomedical databases, encompassing phenotyping and genomic data, has occurred globally (Zhou et al. 2022). Numerous genome-wide association studies (GWAS) have been conducted to determine the genetic architecture underlying a wide range of complex traits and clinical outcomes, with the aim of improving disease prevention and treatment (Uffelmann et al. 2021). Publicly available GWAS summary-level data (or GWAS summary statistics) encompass a wide range of phenotypes. For example, GWAS Catalog (Sollis et al. 2023) is curating and harmonizing GWAS summary statistics for over 90,000 phenotypes. These summary statistics, derived from large-scale studies, provide valuable opportunities for in-depth investigations into genetic overlaps and shared architectures between phenotypes across studies and cohorts. Various statistical genetic tools have been developed to analyze GWAS summary statistics and examine the shared genetic components between pairs of phenotypes, such as LDSC (Bulik-Sullivan et al. 2015), LAVA (Werme et al. 2022), SumHer (Speed and Balding 2019), and Popcorn (Brown et al. 2016). These methods offer insights into genetic links from various perspectives and have been widely applied to clinical biomarkers and outcomes (Lee et al. 2019; Romero et al. 2022).

However, batch-running these tools with the rapidly growing GWAS summary statistics often requires substantial computing efforts and resources due to the lengthy time needed for collecting and downloading large-scale datasets, the high computational demands for data harmonization and preprocessing, and the implementation of diverse tools required for different analyses across all datasets. Consequently, systematic bivariate cross-trait analyses using massive GWAS summary statistics for thousands of phenotypes can be logistically and computationally challenging. As more complex and deep phenotyping data are obtained and disparities in computing resources among researchers grow, addressing these limitations becomes increasingly urgent.

For example, the UK Biobank (UKB) imaging study (Littlejohns et al. 2020) collected multimodal brain imaging data, generating over 5000 imaging-derived phenotypes using different imaging modalities and processing pipelines (Elliott et al. 2018; Zhao et al. 2019, 2021; Smith et al. 2021; Zhao et al. 2022). Researchers interested in a specific disease and its genetic connections with imaging biomarkers have traditionally downloaded all the GWAS summary statistics for over 5000 imaging biomarkers from the Oxford BIG Project (*http://big.stats.ox.ac.uk*) and the BIG-KP project (*https://bigkp.org/*), and run their statistical tools in local clusters, which can be inefficient. Such

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challenges are also present in centralized GWAS databases, such as the GWAS Catalog (Sollis et al. 2023) and IEU OpenGWAS (Elsworth et al. 2020), where users are expected to download and manage large datasets locally to conduct most analyses.

Several cloud-based research platforms have been developed for genetic and genomic research, though most focus on a single study, univariate trait analysis, or a single analysis method/function. For example, UK Biobank (*https://ukbiobank.dnanexus.com/*) and All of Us (*https://www.researchallofus.org/*) have established cloud computing platforms dedicated exclusively to their own individual-level study data. FUMA (Watanabe et al. 2017) is a hub for univariate trait post-GWAS analysis, allowing exploration of the genomic architecture of a single trait (e.g., identifying genes associated with human height and their genomic profiles). Similarly, Locus Compare (Liu et al. 2019) investigates the relationship between one trait and gene expression data (e.g., genes whose expression levels are associated with height). LD Hub (Zheng et al. 2017) was developed for cross-trait analysis, but it supports fewer than 200 traits and only one method (LDSC from Bulik-Sullivan et al. 2015). Additionally, the LD Hub web server has been offline in recent years, preventing any analysis on that platform. Therefore, while cloud computing is promising for genetics and genomics, existing web tools are not well-suited for cross-trait genetic analysis with GWAS summary statistics, and no comprehensive platform currently integrates multiple methods and emerging data resources for these tasks.

To address these limitations, we developed BIGA (*https://bigagwas.org/*), an online cloud-based platform that offers unified data harmonization and analysis pipelines and processed data resources for cross-trait analyses using GWAS summary statistics (Figure 1). BIGA aims to provide various tools for quantifying cross-trait genetic architectures, such as genome-wide genetic correlation methods (e.g., LDSC from Bulik-Sullivan et al. (2015), Popcorn from Brown et al. (2016), and SumHer from Speed and Balding (2019)) and local genetic correlation analysis (e.g., LAVA from Werme et al. (2022)). We have also aggregated and harmonized GWAS summary statistics from various resources, including the GWAS Catalog (Sollis et al. 2023), UKB study (Neale Lab) (Bycroft et al. 2018), Psychiatric Genomics Consortium (PGC) (Sullivan 2009), FinnGen (Kurki et al. 2023), Biobank Japan (Sakaue et al. 2021), CHIMGEN (Xu et al. 2020), UKB-PPP (Sun et al. 2023), BIG-KP (Zhao et al. 2021, 2019, 2022), and Oxford BIG40 (Elliott et al. 2018; Smith et al. 2021). These curated datasets, currently including over 15,000 traits, have been integrated with multiple methods, facilitating easy online analysis for users. In addition, we developed a novel online query function that provides direct access to popular genetic databases, including over 30,000 harmonized datasets in the GWAS Catalog. With our established infrastructure in place, we are committed to the continuous development and growth of BIGA, aiming to broaden its capabilities by consistently including new tools and data resources.

2. BIGA Data Analysis Pipeline Platform

2.1. BIGA Functionality

BIGA's functionality is divided into two main modules: Pairwise Analysis and Massive Analysis. In the Pairwise Analysis module, users can input two sets of summary statistics to perform bivariate analysis on a specific pair (e.g., depression and total brain volume). In the Massive Analysis module, users can input one set of summary statistics (e.g., depression) and select from a wide range of curated datasets in BIGA to screen for associations across multiple phenotypes (e.g., all brain and organ imaging

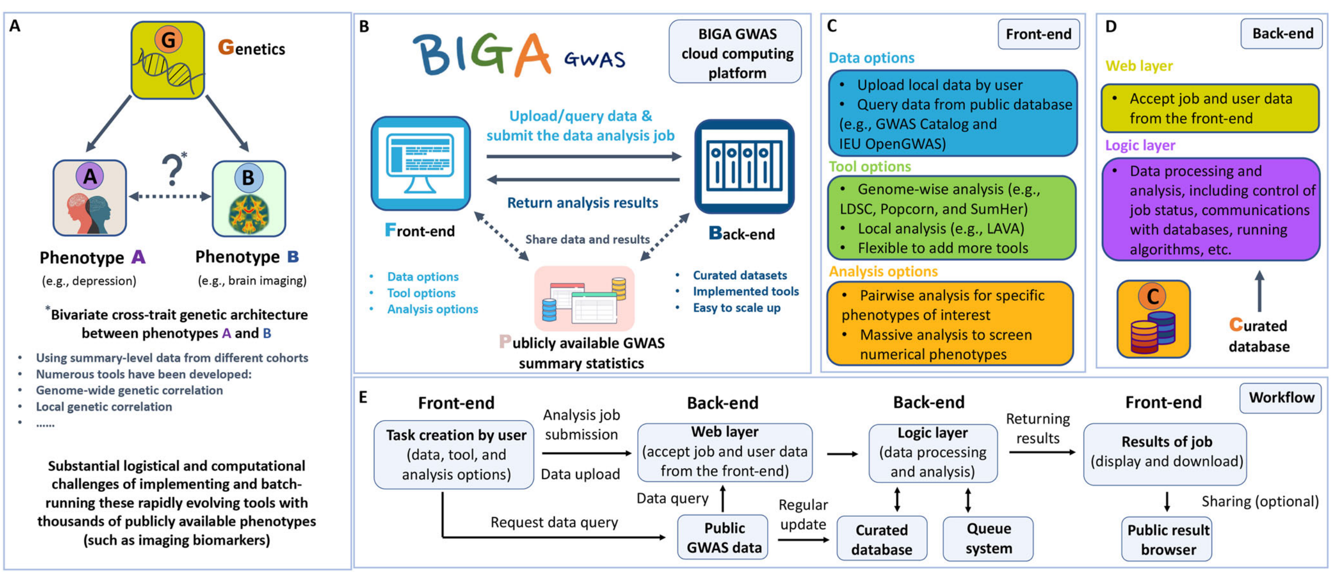

Figure 1. Overview of BIGA GWAS cloud computing platform. (A) The motivation of this project is to address the substantial logistical and computational challenges associated with implementing and batch-running the constantly evolving tools for cross-trait genetic architecture analysis. Our aim is to offer a cloud computing-based solution that can effectively overcome these challenges. (B) Overview of the BIGA GWAS platform. Users can easily upload or query GWAS summary statistics and submit data analysis jobs through the front-end interface. These jobs are then processed on the back-end, and the results are subsequently returned to the users. (C) The front-end interface of the BIGA GWAS platform offers users a comprehensive set of options to manage their data resources, choose the appropriate tools, and select the desired mode of data analysis. (D) Details of the back-end of the BIGA GWAS platform. (E) Overview of the analysis workflow.

traits). To enable users to perform efficient screening across different data resources, we have curated and harmonized over 15,000 phenotypes from nine publicly available GWAS summary statistics resources of European and Asian populations (supplementary Note 1). These include psychiatric disorders from PGC (Sullivan 2009), clinical outcomes from FinnGen (Kurki et al. 2023), plasma proteins from UKB-PPP (Sun et al. 2023), deep phenotypes from the UKB study (Neale Lab) (Bycroft et al. 2018) and Biobank Japan (Sakaue et al. 2021), brain and organ imaging traits from Oxford BIG40 (Elliott et al. 2018; Smith et al. 2021), BIG-KP (Zhao et al. 2019, 2021, 2022), and CHIMGEN (Xu et al. 2020), as well as selected harmonized traits from the GWAS Catalog. These datasets allow users to conduct large-scale bivariate cross-trait analyses efficiently without the need for additional data preprocessing.

2.2. Data Input Options

We offer three options for user-input GWAS summary statistics: "Upload data", "Query data", and "Use previous data", with corresponding data processing steps tailored for each option (Figure 2). We have implemented a unified data harmonization pipeline for user-uploaded data (through the "Upload data" option) (supplementary Note 2). Specifically, BIGA supports user uploads of GWAS summary statistics in either .txt or .gz formats, with file sizes up to 600MB. Files can be structured using either space or tab separation. BIGA's integrated liftover tool allows for the processing of GWAS summary statistics from different genome builds, including GRCh37 and GRCh38. Since the data analysis methods in BIGA, such as LDSC and SumHer, are designed to work with summary statistics from specific populations, users are required to specify the population when uploading GWAS summary statistics.

Upon upload, each file undergoes a rigorous quality check, such as column verification and missing value detection. For data that passes these quality controls (QCs), BIGA will further

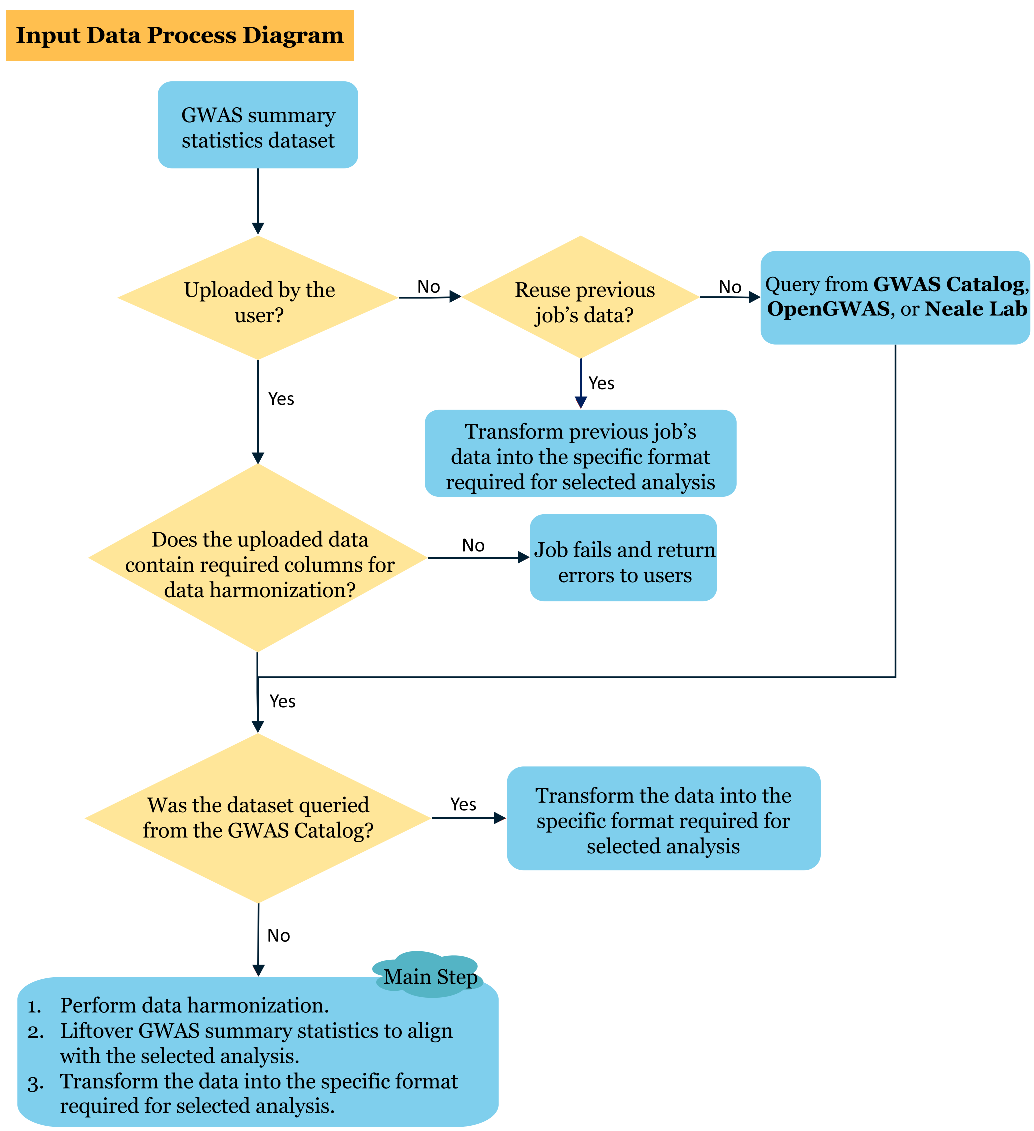

Figure 2. Input data processing diagram. This figure illustrates how BIGA processes different types of input data. We provide three options for input GWAS summary statistics: "Upload data", "Query data", and "Use previous data", with corresponding data processing steps tailored for each option.

harmonize it into a standardized format (Table S1), making it ready for use with all analytical methods implemented in BIGA. Following the GWAS Catalog protocol, our harmonization steps include mapping each variant ID to its genomic location, inferring the orientation of palindromic variants, harmonizing each variant with the Ensembl VCF reference, and filtering out invalid records through QCs. If a job fails due to incorrect input data, BIGA will provide a clear and detailed error message to help users resolve the issue (supplementary Note 3).

In addition, our unified data management system and standardized data format enable the reuse of harmonized data across different methods implemented in BIGA. For example, after completing an LDSC job and identifying a significant genome-wide genetic correlation between two traits, users can further analyze local genetic correlation patterns in each genomic region using LAVA. To facilitate this, users can access previously uploaded and harmonized data via the "Use previous data" option, eliminating the need to re-upload and re-harmonize the data. Users can also download the harmonized input data from BIGA after the completion of their jobs for further use.

2.3. Uploading GWAS Summary Statistics

In this section, we outline the requirements for user-uploaded data in more detail. To ensure data harmonization and compatibility with analysis tools, user-uploaded data must include specific columns as outlined in Table S1. BIGA automatically detects common column names using rules similar to those in the LDSC munge file. For example, BIGA recognizes variations of column names such as "Position", "POS", "Base_pair_location", and "BP" as the "position" field, referring to the base pair position of the single nucleotide polymorphism (SNP) on the chromosome.

Users need to specify the human genome build version (GRCh37 or GRCh38). If the input data includes an SNP column (rsID), BIGA will retrieve chromosome and GRCh38 positions by matching with dbSNP build 156. Additionally, users are required to specify the population of the summary statistics (e.g., European). Effect allele frequency (EAF) is optional, though recommended for analyses involving minor allele frequency cutoffs. Users can also manually input nonstandard column names in BIGA's "Build input files" panel to ensure accurate column mapping. We retain user-uploaded datasets privately for a maximum of seven days before automatic deletion. More details are available in the online documentation: https://bigagwas.org/documentation.

2.4. Querying GWAS Summary Statistics

Another useful data-input option offered by BIGA is the "Query data" function, which enables users to directly access GWAS summary statistics from existing data resources. BIGA has integrated three well-organized summary statistics resources to facilitate easy data access. Specifically, users can query data directly from the GWAS Catalog, IEU OpenGWAS, and UKB study (Neale Lab) without needing to download and re-upload the data to BIGA. We have implemented different strategies for querying data, tailored to the specific features of each resource. For example, we have prepared a daily updated list of queryable GWAS Catalog data. Users can simply input the Trait ID, sample size, and population information, then BIGA will handle the data download and following-up data analysis. The query function enables users to quickly access their desired traits without the need to download, format, or harmonize the data themselves.

For GWAS Catalog, BIGA focuses on their harmonized datasets. During internal testing, we observed variability in the column names of a small portion of these files. For example, the "beta" column may be labeled as "hm_beta" or simply "beta". In such cases, BIGA first searches for "hm_beta" and changes to "beta" if "hm_beta" is absent or contains a large proportion of missing values (e.g., over 50%). Additionally, some early GWAS Catalog harmonized data have column names that don't align with their recent standards.

To minimize unsuccessful queries and ensure only reliable data are processed for downstream analyses, we pre-queried all available GWAS Catalog harmonized datasets and created a list of queryable GWAS data for BIGA analysis. Briefly, we screened each dataset for missing values, performed text mining on each trait's webpage to gather detailed information, and finalized a list that includes trait ID, description, ancestry, sample size, SNP count, HapMap3 SNP count, and column names without excessive missing values. Additionally, we have implemented a periodic job in BIGA to update this list daily, ensuring it remains aligned with the most recent version of the GWAS Catalog harmonized datasets.

After confirming that the data of interest is on the queryable list, users only need to input the Trait ID (GWAS study ID), sample size, and population information. BIGA handles the FTP query and data download. Since the dataset is pre-harmonized, BIGA bypasses the harmonization step and directly converts the data into BIGA's standardized format, which includes columns such as snp, chr, position, pvalue, eaf, a1, a2, n, odds_ratio or beta, and se (Table S1).

For data queried from IEU OpenGWAS, users first need to confirm the trait is queryable by checking if the "Download VCF" button exists on the IEU OpenGWAS trait website. To proceed, users need to input the Trait ID (GWAS ID), sample size, genome-build version, and population, all of which can be found on the IEU OpenGWAS website. BIGA then downloads and humanizes the original summary statistics into its standardized format, interpreting "ID" as "snp", "CHROM" as "chr", "POS" as "position", "LP" as "negative log p-value", "AF1" as "eaf", "ALT" as "a1", "REF" as "a2", "ES" as "beta", and "SE" as "se", making the data ready for use with the implemented analysis tools.

Finally, for data from the UKB study (Neale Lab), we have pre-downloaded and harmonized the entire dataset. As a result, BIGA directly uses our harmonized dataset for querying input, eliminating the need for further harmonization. Detailed documentation and quick-start examples (*https://bigagwas.org/quickstart*) for using the query data functionality are provided on the BIGA platform.

2.5. Data Harmonization

We have implemented a rigorous harmonization process for user-input data, following the protocol of GWAS Catalog (https://www.ebi.ac.uk/gwas/docs/methods/summary-statistics, Section "Harmonised summary statistics data"). Briefly, whenever a user initiates a job, BIGA harmonizes their uploaded or queried data, unless the summary statistics have been directly queried from the pre-harmonized GWAS Catalog database. We require users to include necessary columns in their input data to ensure successful harmonization; failure to do so will result in job termination accompanied by an appropriate error message, which is recorded in the BIGA log file. Our harmonization pipeline has four key components: (i) Mapping variant IDs to locations, either by rsID or base pair location (if rsID is not provided); (ii) Inferring the orientation of palindromic variants; (iii) Variant harmonization with Ensembl VCF reference; and (iv) Filtering and QCing invalid records. A flowchart of our harmonization pipeline is provided in Figure S1, and a detailed explanation of the harmonization procedure is provided in supplementary Note 4. Following the GWAS Catalog's approach, BIGA will provide a harmonization code for each variant in the harmonized dataset, which is available for downloading after the job is completed.

2.6. Data Analysis and Results

After preparing input data, users can select the data analysis modules, methods, and submit their job. In both Pairwise Analysis and Massive Analysis modules, BIGA integrates several widely used methods for different types of analyses, including LDSC (Bulik-Sullivan et al. 2015) and SumHer (Speed and Balding 2019) for cross-trait genetic correlation analysis, LAVA (Werme et al. 2022) for local genetic correlation analysis, and Popcorn (Brown et al. 2016) for cross-population genetic correlation analysis (supplementary Note 5).

Based on the specific situations of each method, we aim to provide flexible implementations that allow users to select the most appropriate configurations for their studies. For example, in LDSC pairwise analysis, users can choose the option to convert to a liability scale for binary traits. In LAVA, users can set p-value thresholds and confidence interval options. In SumHer, users can specify a phenotype variance cutoff and opt for multiplicative inflation of test statistics. In Popcorn, users can select estimators such as "Regression" or "MLE" and choose between "Genetic effect" and "Genetic impact" correlation models.

Once submitted, the job will be processed in the back-end and executed on our cloud computing platform using the specified datasets and tools. Upon completion, users will receive an email notification, and the results are accessible through the front-end interface. BIGA's "Results Page" provides an interactive table interface to access and manage results, enabling users to quickly identify top-ranking significant genetic correlations, search for specific traits, and download all results. Additionally, BIGA offers a "Public Results" Center, allowing users to share their analysis results (supplementary Note 6 and Figure S2). Job log files are also available for users to review the details of their jobs. A quick-start tutorial and comprehensive documentation are provided on the BIGA website to provide clear guidance and ensure transparency throughout the data analysis process (supplementary Note 7 and Figure S3).

2.7. Comparison with Existing Platforms

We compared BIGA with three existing platforms, LD Hub (Zheng et al. 2017), FUMA (Watanabe et al. 2017), and IEU OpenGWAS (Elsworth et al. 2020), across three dimensions: (i) data processing pipelines and resources, (ii) data analysis capabilities, and (iii) computational efficiency (Table 1). Briefly,

Table 1. Feature comparison among BIGA, LD Hub, FUMA, and IEU OpenGWAS platforms. This table summarizes the features provided by BIGA in comparison to those offered by other existing platforms.

	Features	BIGA	LD Hub	FUMA	IEU OpenGWAS
Data processing pipelines and resources	Data preprocessing	✓	✓	✓	✓
	Unified harmonization and quality control pipelines	✓	×	×	×
	Number of curated datasets	> 15,000	173	—	50,044
	Querying data from external data resources	GWAS Catalog (> 30,000), IEU OpenGWAS (> 38,000), and UK Biobank (> 4,000)	×	×	×
Data analysis pipeline	Cross-population analysis support	✓	×	×	×
	LDSC analysis	✓	✓	×	×
	SumHer analysis	✓	×	×	×
	LAVA analysis	✓	×	×	×
	Popcorn analysis	✓	×	×	×
Computational efficiency	Batch analysis	✓	×	×	×
	Could scalability	✓ (cloud-native, elastic)	×	✓ (cloud-native, elastic)	× (no cloud functionality)
	Job queuing system	✓	×	✓	×

Notation: "✓" indicates supported; "×" indicates not supported; "—" indicates not applicable or not provided.

LD Hub is a web-based tool designed specifically for LDSC analysis but has been offline in recent years. While it offers 173 traits for selection, BIGA provides access to over 15,000 curated traits. FUMA supports functional mapping and annotation for single-trait analysis but lacks support for cross-trait analysis. IEU OpenGWAS hosts a large collection of GWAS summary statistics and provides offline analysis tools; however, it does not provide a cloud-based platform with features such as a job queuing system or batch processing.

In contrast, BIGA introduces a novel data query function that provides direct access to several major GWAS databases, including over 30,000 datasets from the GWAS Catalog (Sollis et al. 2023), over 38,000 from IEU OpenGWAS (Elsworth et al. 2020), and more than 4000 traits from the UKB study (Neale Lab) (Bycroft et al. 2018). BIGA also provides a broader suite of analytical tools supporting both within-population and cross-population genetic analyses. Its harmonization workflows and unified data management framework ensure consistent and reliable data processing. By removing the need to manually download, process, and locally analyze GWAS summary statistics, BIGA significantly reduces the computational burden on users.

3. Code Infrastructure and Validation

3.1. Computational Efficiency and Scalability

To ensure efficient and scalable large-scale data processing and analysis, BIGA is built on a robust and optimized code infrastructure. Every step, from the initial data input to the final results output, is organized by a standardized pipeline, offering the flexibility to incorporate new methods. For example, BIGA is built on the Django 3.2 web framework (*https://www.djangoproject.com/*) and is organized into two main sections: the front-end and the back-end. The back-end handles job creation, data querying, harmonization, and analysis execution. To optimize efficiency, it is divided into six specialized modules:

- A core module for distributing user requests and collecting results;
- A data harmonization module;
- A job management and queuing system;
- A database management module;
- A module for front-end files; and
- A module for storing various method-specific shell scripts.

Each module is responsible for a different function. For example, the core module manages user requests by breaking them down into steps such as job creation, data querying, harmonization, and analysis execution. It distributes these tasks to the appropriate modules and ensures smooth processing, returning the analysis results to the users.

BIGA's infrastructure is designed for high computational efficiency, currently processing six jobs simultaneously with just 32 GB of RAM and 8 Intel vCPUs. BIGA uses Celery (*https://docs.celeryq.dev*) to manage the queuing system and Redis (*https://redis.io/*) to manage message communication. Our setup includes three queues based on different CPU and time usage: one dedicated to less CPU-intensive tasks such as LDSC and SumHer analyses, another for more demanding tasks like LAVA and Popcorn analyses, and a third for routine or quick tasks, including data cleanup and email sending. Techniques such as chunking for data processing and parallel job execution are used to optimize CPU usage. For example, to ensure the stability of our platform and to reduce the memory usage for each job, especially for data harmonization, we process summary statistics using chunking and multiprocessing. In BIGA, the segmented approach allows us to process summary statistics with small memory usage (around 5 GB) for a single operation. This results in a substantial reduction of CPU usage compared to the 28 GB required by the GWAS Catalog harmonization pipeline (*https://github.com/EBISPOT/gwas-sumstats-harmoniser*), which processes the entire data at once.

We conducted large-scale quantitative tests and comparisons to validate the stability and computational efficiency of BIGA. Specifically, we compared BIGA's performance with two powerful university-wide computing clusters: Purdue Bell and UNC Longleaf. For this evaluation, we used 101 imaging traits to perform a batch LDSC analysis, using two datasets: the "PGC dataset" with 24 traits and the "Brain and Organ Imaging Traits" dataset with 360 imaging traits. To ensure a fair comparison, CPU memory was standardized to 16 GB across all platforms. The results showed that while BIGA's average execution time was slightly longer than the other clusters (2% and 7% longer than the best performances), its stability in execution time outperformed the others (Figure S4). It is also worth noting that, with BIGA, users can run jobs automatically with just a few clicks, selecting options directly on the platform. In contrast, running analyses on local clusters requires completing the entire process manually—from downloading and preprocessing data to implementing methods and summarizing results—assuming access to the clusters and sufficient storage space is already available. Additionally, we evaluated the platform's performance under heavy load with a stress test of 50 simultaneous submissions, consisting of 33 pairwise analyses and 17 massive analysis jobs. These jobs involved querying, uploading, and reusing previous data. Our system completed all tasks without any CPU or RAM issues (Figure S5). In summary, these quantitative results show that BIGA offers substantial practical convenience while providing computational efficiency comparable to high-performance advanced clusters, with stable performance under heavy demand.

3.2. Technical Validation of Analysis Results

To assess the technical validity of BIGA's implementation of analytical methods, we compared its outputs with those from offline command-line runs across all supported methods, including LDSC, SumHer, LAVA, and Popcorn. We selected three representative traits from the UKB study: body mass index (BMI, Field 21001), high blood pressure (Field 6150), and stroke (Field 6150). For within-population analyses (LDSC, SumHer, and LAVA), we estimated genetic correlations for the following trait pairs: (i) BMI and 24 traits from the "PGC dataset", (ii) high blood pressure and 82 cardiac imaging traits from the "Brain and Organ Imaging Traits" dataset, and (iii) stroke and 101 regional brain volume traits from the same imaging dataset. For cross-population analyses using Popcorn, we assessed genetic correlations between BMI and 177 traits from Biobank Japan, as well as between both high blood pressure and stroke and 44 traits from CHIMGEN. In all cases, we processed identical input files using both BIGA and the corresponding original standalone pipelines for comparison. Across all methods and trait sets, BIGA's outputs were highly consistent with those from the original command-line tools (Table S2 and Figure S6), demonstrating the accuracy and reliability of the BIGA platform.

4. Application Example in Blood Pressure Analysis

To showcase the extensive genetic analyses that BIGA can conduct, we performed blood pressure data analysis, aiming to explore its complex genetic correlation patterns with different complex traits and diseases available on BIGA. We first queried systolic blood pressure from the IEU OpenGWAS (*https://gwas.mrcieu.ac.uk/datasets/ieu-b-38/*, trait id: ieu-b-38). BIGA performed harmonization for this queried dataset and then used the harmonized data to perform LDSC massive analysis with curated datasets of European ancestry. The curated datasets included 15,428 phenotypes in total: 4347 UKB phenotypes provided by the Neale Lab, 303 clinical outcomes from FinnGen, 3905 brain imaging phenotypes from the Oxford BIG40, 3,012 brain and organ imaging phenotypes from the BIG-KP, 2,940 proteins from UKB-PPP, 897 phenotypes from GWAS Catalog, and 24 traits from PGC and other brain disorder datasets. At a false discovery rate 5% level (P range = $(5.44\times10^{-244}$, $4.00\times10^{-2})$), we found that systolic blood pressure was widely associated with different categories of phenotypes among the curated datasets (Figure S7). Besides well-known associations with diastolic blood pressure and hypertension, we found 293 additional significant associations in UKB (Neale Lab), 64 in FinnGen, 265 in Oxford BIG40, 226 in BIG-KP, 74 in UKB-PPP, and 282 in GWAS Catalog (Table S3). For example, systolic blood pressure had significant genetic correlations with heart disease-related traits (Ischaemic heart disease, Illnesses of mother: Heart disease, Illnesses of father: Heart disease) in UKB (Neale Lab) (correlation = 0.31, 0.38, and 0.29, P = 1.05×10^{-28}, 2.98×10^{-24}, 1.36×10^{-19} respectively), cardiovascular diseases in FinnGen (correlation = 0.53, $P = 4.41\times10^{-136}$), and coronary atherosclerosis in GWAS Catalog (correlation = 0.33, $P = 2.21\times10^{-29}$). As expected, among all imaging traits, the strongest associations were observed on the heart imaging traits (49 were significant traits, $P < 2.30\times10^{-3}$). These traits included regional/global myocardial wall thickness, left ventricular myocardial mass, ascending/descending aorta distensibility, regional/global radial strain, and left/right ventricular cardiac output. Second, we found 437 significantly associated plasma proteins in UKB-PPP, with the strongest genetic correlations being observed on MMP7 (correlation = 0.25, $P = 1.73\times10^{-04}$), PTPRB (correlation = 0.23, $P = 1.58\times10^{-07}$), and SIGLEC7 (correlation = 0.21, $P = 4.95\times10^{-04}$). We also performed genetic correlation analysis with diastolic blood pressure (*https://gwas.mrcieu.ac.uk/datasets/ieu-b-39/*, trait id: ieu-b-39) and found similar association patterns to systolic blood pressure (Figure S8 and Table S4). We applied the SumHer method to repeat these analyses (Figure S9 and Tables S5–S6) and observed that results from LDSC and SumHer were generally consistent (Figure S10, Pearson's correlation = 0.9273).

To further investigate the connection between the two blood pressure measures and cardiovascular diseases, we conducted a local genetic correlation analysis with coronary artery disease (CAD) through BIGA pairwise analysis using LAVA. LAVA identified 86 local genetic regions where CAD showed genetic correlations with systolic blood pressure, with 83 exhibiting positive genetic correlations and 3 showing negative correlations. Similarly, diastolic blood pressure was associated with CAD in 77 regions, 70 of which were positive (Tables S7–S8). Moreover, we explored the cross-population genetic correlation of blood pressure between European and East Asian ancestries. By querying East Asian diastolic and systolic blood pressure traits from the GWAS Catalog (study ID: GCST90278657), we conducted Popcorn analysis using BIGA. The cross-trait genetic correlation estimates were 0.8345 for systolic and 0.7617 for diastolic blood pressure measures, neither of which were significantly different from 1 (P = 0.2538 and 0.1726, respectively, Table S9).

5. Discussion

In this article, we introduced BIGA (*https://bigagwas.org/*), a cloud computing-based web platform that integrates diverse public GWAS summary statistics and cross-trait genetic analysis tools. The primary goal of BIGA is to facilitate large-scale batch genetic analyses by leveraging extensive publicly available GWAS summary data. BIGA offers several key novel features for cross-trait analysis:

(i) Unified file management, simplifying the process of preparing user-input files for immediate use across various tools;
(ii) A rigorous data harmonization pipeline to ensure the high quality of datasets and results. All user-uploaded or queried data, as well as curated datasets, undergo mandatory harmonization unless already processed by GWAS Catalog;
(iii) An efficient query function, enabling users to access desired GWAS summary statistics directly from public resources without needing to download or process the data themselves;
(iv) Multiple methods and curated datasets, allowing users to conduct multiple analyses with data from different resources; and
(v) Efficient CPU usage, using chunking and multiprocessing techniques to significantly reduce the memory requirement of each job.

BIGA can efficiently analyze approximately 10,000 traits within a 24-hr timeframe. To ensure smooth operation, BIGA has a job management queuing system, tracking each job from start to completion. Transparent error messages and log files are also provided for users.

Since the launch of the BIGA website, we have actively monitored user engagement and platform usage (Figure S11). To promote transparency, we introduced a feature that displays key usage metrics, including the total number of users, completed jobs, currently running jobs, and jobs in the queue. We are open to external collaborations to further expand BIGA's functionality, supported methods, data resources, and practical impact. To support a growing GWAS database and an expanding user base, it is essential to ensure that BIGA's infrastructure remains scalable and cost-efficient. Notably, built on a cloud-based architecture, BIGA can dynamically adjust both storage and compute resources as needed (Figure S12). Additional details on user adoption and infrastructure scalability are discussed in supplementary Note 8.

In summary, BIGA provides a new ecosystem that enables researchers to easily perform multiple cross-trait analyses without needing access to local computing clusters, implementing methods locally, or downloading large datasets. By lowering computational barriers for large-scale data analysis with rapidly growing genetic datasets, BIGA will help reduce resource disparities within the research community and attract a broader user base to these developed methods. The source code for building the BIGA platform will be made publicly available on GitHub, facilitating the development of cloud computing tools with similar designs in other research fields. The BIGA website welcomes user feedback and requests, which aids in improving the project and implementing new tools and functions to better meet the needs of the research community.

Supplementary Materials

The supplementary materials include supplementary text, supplementary figures, supplementary tables, and source code.

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Disclosure Statement

The authors report there are no competing interests to declare.

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Data Availability Statement

GWAS summary statistics used in the BIGA platform are publicly available and can be found in several public databases, such as the:

- UK Biobank (Neale Lab): *http://www.nealelab.is/uk-biobank*
- Psychiatric Genomics Consortium: *https://pgc.unc.edu/*
- IEU OpenGWAS: *https://gwas.mrcieu.ac.uk/*
- FinnGen: *https://www.finngen.fi/en/access_results*
- Biobank Japan: *https://pheweb.jp/*
- BIG-KP: *https://bigkp.org/*
- Oxford BIG40: *https://open.win.ox.ac.uk/ukbiobank/big40/*
- UKB-PPP: *https://metabolomips.org/ukbbpgwas/*
- GWAS Catalog: *https://www.ebi.ac.uk/gwas/downloads/summary-statistics*
- CHIMGEN: *http://chimgen.tmu.edu.cn/en/index.php?c=article&id=2036*

Code availability

All source code to develop the BIGA platform is publicly available on the BIGA GitHub repository (*https://github.com/yuj23/bigagwas*). The statistical tools and methods implemented in the BIGA platform are also open source, and their source code has already been made available to the public by their authors. A summary of our implemented tools and data resources can be found at https://bigagwas.org/documentation. The blood pressure example datasets and scripts used in Section 4 are publicly available in *https://zenodo.org/records/15831835*.

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